DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Survival Models: Overview

Learning Objectives

- Describe the contexts of survival analysis
- Explain the differences among duration, survival function, and hazard function
- Construct and estimate Proportional Hazard models

Survival / Duration Analysis

- Survival / Duration Analysis
  - Binary Response + ***Time*** Information
  - **Duration**: the time lapse between when an individual enters an initial state and when that individual exits the initial state
    - An observation is right-censored if it remains in the initial state till the end of the observation window
  - Data structure:
    - Independent variables as usual
    - Dependent variable consists of *two* pieces of information:
      - (i) whether an individual observation is right-censored or not;
      - (ii) the time lapse till an individual exits the initial state or gets right-censored

Survival / Duration Analysis (cont'd)

- Survival / Duration Analysis
  - Examples:
    - How long patients survive after a treatment
    - How long a company remain in business before going bankrupt
    - How long a machine can operate properly before going out of order
    - How long people remain unemployed before finding another job
    - How long customers stay with the company before "churn"
    - ……
  - Interested in what variables could affect:
    - *Expected* duration
    - ***Hazard***: the instantaneous probability of exiting the current state at any given time

Survival Analysis Basics

- Duration T:
  - The time lapse till when an individual exits the initial state since s/he enters the state
  - Fully observed for uncensored observations; partially observed for censored observations
  - T is a random variable
  - cdf: $F(t) = \Pr(T \leq t)$ pdf: $f(t) = \frac{dF}{dt}(t)$
- Survival Function

$$S(t) = 1 - F(t) = \Pr(T > t)$$

- Hazard Function

$$\lambda(t) = \lim_{h \to 0} \frac{\Pr(t \leq T < t + h | T \geq t)}{h}$$

$$\Pr(t \leq T < t + \Delta h | T \geq t) = \lambda(t) \cdot \Delta h$$

  - $\lambda(t)$ is the instantaneous probability density of an individual exiting the current state at time t, *conditional on* that s/he has survived up to time t.

Hazard Function

- Equivalence of duration distribution, survival, and hazard functions:

$$\lambda(t) = \lim_{h \to 0} \frac{F(t+h) - F(t)}{h} \cdot \frac{1}{1 - F(t)} = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$$

$$\lambda(t) = -\frac{d \ln S(t)}{dt}$$

$$S(t) = \exp[-\int_0^t \lambda(s)\, ds]$$

$$F(t) = 1 - \exp[-\int_0^t \lambda(s)\, ds]$$

$$f(t) = \lambda(t) \exp[-\int_0^t \lambda(s)\, ds]$$

- Simplest hazard function:
  - $\lambda(t) \equiv \lambda$ constant hazard
  - *Memoryless*: probability of exiting doesn't depend on time
  - Duration T follows *exponential* distribution: $f(t) = \lambda e^{-\lambda t}$, $F(t) = 1 - e^{-\lambda t}$

Proportional Hazard Models

- Consider time-invariant covariates
- **Proportional Hazard (PH) Model**

$$\lambda(t; X) = \kappa(X) \cdot \lambda_0(t)$$

  - $\kappa(X) > 0$: a positive function of covariates X; Commonly, $\kappa(X) = e^{X\beta}$
  - $\lambda_0(t) > 0$: **baseline hazard**

- Likelihood function

$$\ell(t_i|X_i, d_i) = f(t_i|X_i)^{d_i}[1 - F(t_i|X_i)]^{(1-d_i)}$$

  - $d_i = 1$ uncensored; $d_i = 0$ censored;

- Log-likelihood function for the whole sample:

$$\mathcal{L}(\beta, \theta) = \sum_i \{d_i \ln[f(t_i|X_i)] + (1 - d_i) \ln[1 - F(t_i|X_i)]\}$$

  - θ: parameters in the baseline hazard function
  - MLE estimation

Summary

- Survival analysis basics:
  - Duration
  - Survival function
  - Hazard function
- Proportional Hazard models:
  - Setup
  - Estimation

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Survival Models: Weibull Proportional Hazard

Learning Objectives

- Describe the Weibull distribution
- Construct and estimate Weibull Proportional Hazard models
- Interpret Proportional Hazard model coefficients

Weibull Distribution

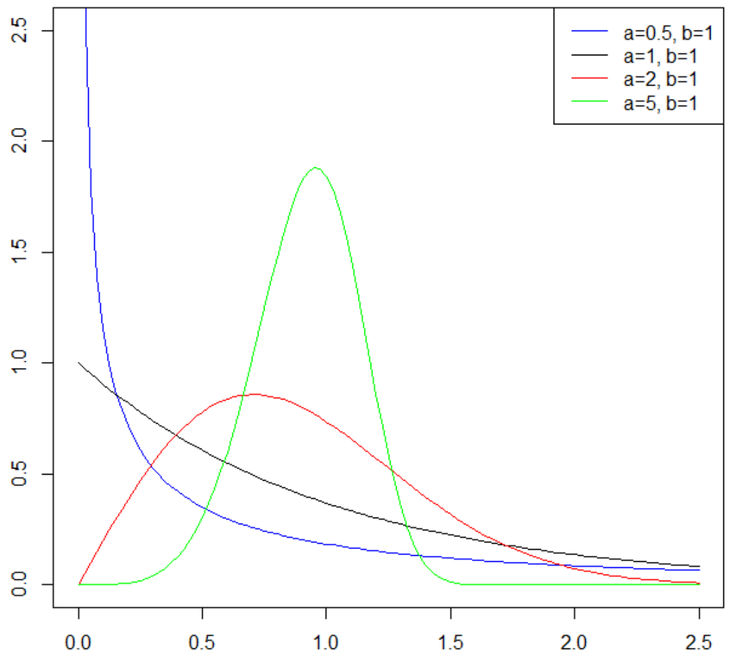
- The Weibull distribution -- for Duration T (>0)
- Different versions of parameterization:
  - $f(t) = (\frac{a}{b})(\frac{t}{b})^{a-1}\exp(-(\frac{t}{b})^{a})$ standard version used in R
  - $f(t) = \gamma\alpha t^{\alpha-1}\exp(-\gamma \cdot t^{\alpha})$ commonly used in econometrics
  - $f(t) = \lambda^{\alpha}\alpha t^{\alpha-1}\exp(-\lambda^{\alpha} \cdot t^{\alpha})$ another common version

- Follow the standard parameterization

$$f(t) = b^{-a} \cdot a \cdot t^{a-1} \cdot \exp(-b^{-a} \cdot t^{a})$$
$$F(t) = 1 - \exp(-b^{-a} \cdot t^{a})$$
$$\lambda(t) = b^{-a} \cdot a \cdot t^{a-1}$$

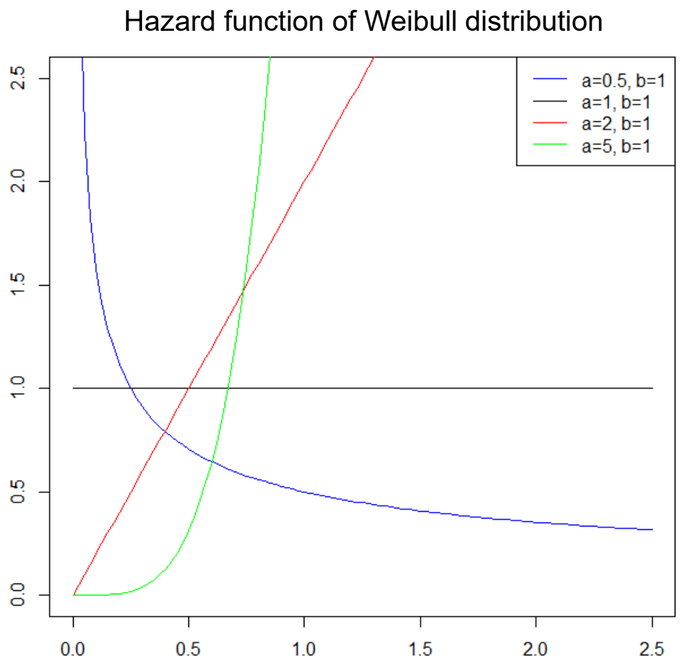
  - $a\ (> 0)$: shape parameter
  - $b\ (> 0)$: scale parameter

pdf of Weibull distribution



Weibull Distribution (Cont'd)

- The Weibull distribution: $\lambda(t) = b^{-a} \cdot a \cdot t^{a-1}$
  - When $a = 1$, $\lambda(t) \equiv b^{-1}$: hazard is constant
    - Weibull distribution reduces to *exponential* distribution
  - When $a > 1$, $\lambda(t)$ is increasing in t, **positive duration dependence**
  - When $a < 1$, $\lambda(t)$ is decreasing in t, **negative duration dependence**



- $S(t) = \exp(-b^{-a} \cdot t^{a})$

  $\ln(-\ln S(t)) = a \ln(t) - a \ln b$
- Log-log of survival function is *linear* in log of time
- <u>Examine whether $\ln(-\ln S(t))$ and $\ln(t)$ fit to a straight line</u>

Weibull PH Model

- Weibull Proportional Hazard Model with covariates:

$$\lambda(t; X) = e^{X\beta} \cdot a \cdot t^{a-1}$$

  - Baseline hazard $\lambda_0(t) = a \cdot t^{a-1}$ is a Weibull hazard (with $b = 1$)
  - Full hazard $\lambda(t; X)$ is also a Weibull hazard (with $\boldsymbol{b = e^{-\frac{X\beta}{a}}}$)
  - *Weibull distribution is closed under proportional hazard specification*

- pdf and cdf:

$$f(t) = e^{X\beta} a t^{a-1} \cdot \exp(-e^{X\beta} t^{a})$$

$$F(t) = 1 - \exp(-e^{X\beta} t^{a})$$

- Log-likelihood:

$$\mathcal{L}(\beta, a) = \sum_i \{d_i \ln[f(t_i|X_i)] + (1 - d_i) \ln[1 - F(t_i|X_i)]\}$$

Interpreting Coefficients

- Interpreting β:

$$\ln \lambda(t; X) = X\beta + \ln \lambda_0(t)$$

$$\beta_j = \frac{\partial}{\partial x_j} \ln \lambda(t; X) = \frac{\partial \lambda(t; X)}{\partial x_j} \cdot \frac{1}{\lambda(t; X)}$$

  - β_j measures the *semi-elasticity* of x_j on the ***hazard***
  - Unit change in x_j changes the *hazard* by (roughly) $100\beta_j$ percent
    - E.g., $\beta_j = 0.25$ means if x_j increases by one unit, the hazard at any time t, $\lambda(t)$, will increase by roughly 25%

Alternative Weibull Model Specification

- <u>Weibull model in regression form</u>
  - For the duration T following the Weibull PH model, it can be shown:

    $\varepsilon = -a\ln(T) - X\beta$ follows *type-I extreme value distribution*
  - Weibull PH model can thus be presented in a regression form:

$$\ln(T) = X\delta + e$$

    where $\boldsymbol{\delta_j = -\frac{\beta_j}{a}}$, and $e = -\frac{1}{a}\varepsilon$ (i.e., a separate error term)
  - δ_j measures the *semi-elasticity* of x_j on the ***expected duration***
  - Unit change in x_j changes the expected duration by (roughly) $100\delta_j$ percent

R Functions

- **survival** package
- **survreg**(): survival regression, with dist="weibull"
  - **Note**: coefficient estimates reported from **survreg**() function are $\boldsymbol{\delta}$, not $\boldsymbol{\beta}$!!
  - The coefficient estimates need to be transformed as follows:

$$\text{Weibull shape parameter } \boldsymbol{a} = \frac{\mathbf{1}}{\boldsymbol{scale}}$$

$$\boldsymbol{\beta} = -\boldsymbol{\delta} \cdot \boldsymbol{a}$$

- **Surv**(time, event, …): create a survival object for analysis
  - time: duration times
  - event: 0=right-censored, 1=not censored (Often opposite to how "censored" is coded)

- **dweibull**(x, shape=a, scale=b): Weibull pdf

Summary

- Weibull distribution
  - pdf, cdf, hazard
  - Positive/negative duration dependence
- Weibull Proportional Hazard model
  - Setup
  - Estimation
- Interpret Proportional Hazard model coefficients
  - Semi-elasticity of x_j on the hazard